Decoupled Amplitude and Phase Recalibration for Tiny Object Detection in Remote Sensing

Kai Zheng, Jintao Li, Feng Sun, Lin Li, Shui Yu, *Member, IEEE*, and Yun Li, *Fellow, IEEE*

***Abstract*—Tiny object detection in remote sensing imagery remains difficult because repeated convolutional downsampling progressively suppresses structurally informative frequency responses, which weakens fine-grained cues and degrades localization accuracy. To address this issue, we present a frequency-aware detection framework built on a YOLO11-style one-stage detector. We first analyze the frequency characteristics of tiny-object representation and show that backbone downsampling increasingly shifts features toward low-frequency responses, while phase plays a more direct role than amplitude alone in preserving tiny-object structure. Motivated by this observation, we propose a frequency-aware amplitude and phase recalibration module (FAPR), which transforms intermediate features into the frequency domain and performs frequency-conditioned decoupled modulation on amplitude and phase. This module provides a compact learnable mechanism for compensating for the systematic spectral bias introduced by repeated downsampling. Lightweight context aggregation and head-level refinement are further incorporated as complementary components. Experiments on AI-TOD, DIOR, and USOD demonstrate consistent improvements over strong baselines, achieving 28.7% AP, 48.5% AP, and 36.10% AP, respectively. Further analysis and ablation results verify the effectiveness of the proposed frequency-aware design, especially the benefit of decoupled amplitude-phase compensation for tiny-object representation.**

***Index Terms*—Fourier transform, frequency domain, remote sensing, small object detection.**

I. Introduction

REMOTE sensing object detection plays an important role in environmental monitoring, urban planning, disaster assessment, and security-related applications [1], [2]. Although deep learning-based detectors have achieved substantial progress, accurately detecting tiny objects in high-resolution remote sensing imagery remains difficult. Such targets often occupy only a few pixels and usually appear under dense layouts, severe scale variation, arbitrary orientations, and cluttered backgrounds [3]. Under these conditions, the discriminative structures required for recognition and localization are easily weakened during hierarchical feature extraction, which often leads to missed detections and degraded localization accuracy [4], [5].

A central challenge is that the difficulty of tiny-object detection does not arise only from object size itself, but also from the progressive weakening of structural cues as features propagate through the backbone. Existing detectors have improved performance through stronger feature extraction, context aggregation [6], [7], and cross-scale interaction [8], [9]. However, most of these methods mainly enhance detection in the spatial domain and provide limited insight into how repeated downsampling degrades tiny-object-sensitive representations. As a result, the representation weakness of tiny objects is often alleviated indirectly rather than explicitly analyzed and compensated.

A frequency-domain perspective offers a more direct way to examine this degradation [10], [11]. In remote sensing imagery, low-frequency components are often associated with coarse background trends, whereas middle-frequency and high-frequency components are more closely related to edges, textures, and fine structural variations [12]. For tiny objects, these informative cues are already weak and are further attenuated by repeated downsampling. This observation suggests that the difficulty of tiny-object detection is closely related to frequency degradation in the feature hierarchy, rather than to insufficient spatial enhancement alone.

Another important issue is that spectral information should not be treated as a uniform entity [10]. In visual representation, amplitude mainly reflects spectral energy distribution, whereas phase is more directly related to spatial arrangement and structural integrity. This distinction is especially relevant to tiny-object detection, where recognition often relies on weak contours, compact boundaries, and subtle geometric organization. Therefore, a uniform spectral modulation strategy is insufficient for this problem. A more suitable solution should account for the different roles of amplitude and phase, and should be guided by the degradation behavior observed during feature extraction.

Motivated by these observations, we develop a YOLO11-style one-stage detector centered on frequency-aware backbone compensation. The core component is the proposed frequency-aware amplitude and phase recalibration module (FAPR), which transforms intermediate feature maps into the frequency domain and performs frequency-conditioned decoupled modulation on amplitude and phase. In this way, the detector compensates for the systematic spectral bias

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introduced by repeated downsampling while strengthening tiny-object-sensitive structures. In addition, a lightweight dynamic-scale RepViT (DS-RepViT) block is used for context aggregation, and a frequency-guided residual attention module (FRAM) is introduced in the detection head to further refine prediction features.

Extensive experiments on AI-TOD, DIOR, and USOD demonstrate consistent improvements over strong baselines, with especially clear gains on very tiny targets and under stricter localization metrics. In addition to benchmark evaluation, we conduct frequency-oriented analysis, factorial ablation, and internal FAPR studies to verify the proposed design and its underlying motivation. These results suggest that mechanism-aware frequency modeling provides an effective direction for remote sensing of tiny objects.

The main contributions of this work are summarized as follows.

1) We provide a frequency-oriented analysis of tiny-object representation in convolutional backbones. The analysis shows that repeated downsampling progressively biases features toward low-frequency responses while suppressing structurally important components, and further indicates that phase plays a more direct role than amplitude alone in preserving tiny-object structure.
2) Motivated by this analysis, we propose the FAPR. FAPR performs frequency-conditioned decoupled modulation on amplitude and phase, thereby compensating for frequency degradation during feature extraction and strengthening tiny-object-sensitive structures.
3) We integrate lightweight context modeling and prediction-stage frequency-guided refinement as complementary components, and validate the overall design through experiments on three benchmarks together with factorial ablation and internal module analysis.

II. Related Work

A. Remote Sensing Tiny-Object Detection

Remote sensing object detection remains challenging because of extreme scale variation, dense layouts, arbitrary orientations, and cluttered backgrounds [13]. These difficulties are more pronounced for tiny objects, whose discriminative cues often occupy only a few pixels and are therefore easily overwhelmed by background responses [3]. Early studies mainly improved detection robustness through stronger detection pipelines, better label assignment, and refined geometric modeling [14], [15]. More recent tiny-object-oriented detectors have further explored feature alignment, hierarchical activation, contextual enhancement, and reconstruction-based refinement to strengthen weak target cues in aerial scenes [16], [17].

Lightweight YOLO-style designs are also attractive for remote sensing because they offer favorable efficiency on large-area imagery and can be adapted to dense small-object scenarios through tailored heads, shallow feature enhancement, and compact cross-scale interaction [18], [19], [20]. These studies have shown the importance of preserving fine-resolution information and improving local discrimination. However, most existing methods still focus on spatial feature enhancement, while the representation degradation of tiny objects during hierarchical feature extraction remains less explicitly examined.

B. Frequency-Domain Modeling for Visual and Remote Sensing Detection

Frequency-domain modeling provides a complementary perspective to spatial feature learning because Fourier and wavelet representations can expose structural responses that are less explicit in the spatial domain [10], [21]. Existing studies have explored spectral-enhanced backbones, spatial-frequency hybrid representations, and frequency-guided attention to improve structural completeness, detail preservation, and background suppression [11]. Some methods further introduce multiscale spatial-frequency interaction or dual-space enhancement to stabilize representations across stages and domains [22], [23]. In remote sensing imagery, frequency-aware designs have also been used to emphasize informative responses and suppress cluttered interference [24].

Despite this progress, several issues remain for remote sensing tiny-object detection. Many frequency-based methods introduce spectral operations as generic enhancement modules, but provide limited analysis of which frequency components are most affected in tiny-object representation. In addition, most existing studies treat the spectrum as a unified entity, while the distinct roles of amplitude and phase receive much less attention. Recent tiny-object-oriented frequency-aware detectors have shown the value of spectral cues [25], [26], yet the mechanism-level analysis of tiny-object frequency degradation and the explicit decoupling of amplitude and phase remain insufficiently explored.

C. Lightweight Context Modeling and Prediction Refinement

Besides feature preservation, remote sensing tiny-object detection also relies on effective context aggregation. Transformer-based detectors improve long-range dependency modeling and global context reasoning, which is beneficial when local target evidence is weak or spatially scattered [6]. Hybrid convolution–transformer designs further seek a better balance between local detail extraction and large-range interaction in aerial scenes [7], [27]. To reduce computational cost, recent studies have investigated efficient attention, lightweight context modules, and parameter-efficient adaptation for remote sensing perception [28], [29], [30]. Transformer-enhanced tiny-object detectors also indicate that contextual reasoning can improve robustness under dense layouts and cluttered backgrounds [31].

Multiscale aggregation remains another common ingredient in remote sensing detection because tiny objects rely heavily on high-resolution representations and cross-scale information exchange [8]. Representative pyramid designs improve top-down and bottom-up information flow [9], [32], while scale-aware and lightweight variants further enhance fine-resolution features with lower complexity [19], [33]. These studies improve semantic propagation and feature interaction,

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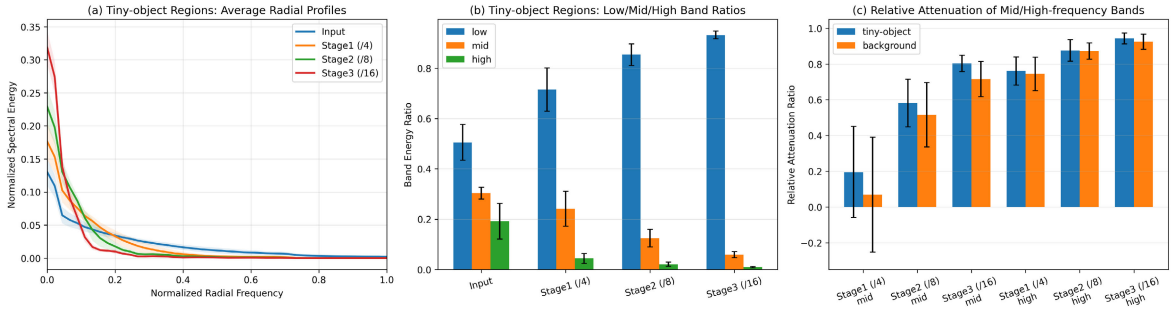


Fig. 1. Spectral statistics of tiny-object regions across backbone stages. (a) Average radial spectral profiles of tiny-object patches from the input image and successive backbone stages. (b) Energy ratios of low-, mid-, and high-frequency bands. (c) Relative attenuation of mid- and high-frequency components for tiny-object regions and background regions.

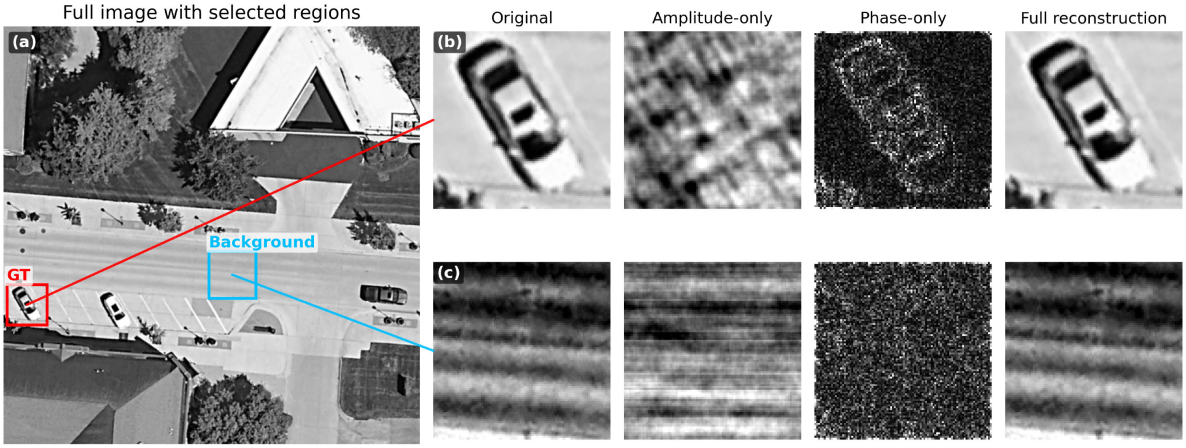


Fig. 2. Amplitude–phase reconstruction analysis. (a) Full remote sensing image with selected tiny-object and background regions. (b) Reconstruction results for the tiny-object patch using the original spectrum, amplitude-only spectrum, phase-only spectrum, and full reconstruction. (c) Reconstruction results for the background patch. Amplitude-only reconstruction preserves coarse intensity tendency but fails to recover spatial structure, whereas phase-only reconstruction retains the structural layout despite intensity distortion.

but they are not designed to directly address the frequency degradation introduced by repeated downsampling. This limitation suggests that context modeling and feature refinement are more suitable as complementary components than as direct solutions to the representation problem itself.

III. Frequency Mechanism of Tiny-Object Representation

Tiny objects in remote sensing imagery typically occupy only a few pixels and often exhibit weak visual contrast against complex backgrounds [34]. When processed by convolutional backbones, repeated downsampling and feature aggregation can progressively suppress the fine-grained structures required for accurate localization [35]. To examine this effect, we analyze tiny-object representation from two complementary perspectives. First, we study how convolutional downsampling reshapes spectral distributions across backbone stages. Second, we examine the respective roles of amplitude and phase in preserving tiny-object structure. These analyses provide a mechanism-level motivation for the frequency-aware design adopted in our method.

A. Downsampling-Induced Frequency Attenuation in Tiny-Object Regions

Tiny objects have limited spatial support and depend heavily on localized structural variations. In standard frequency-domain interpretations, such fine structures are more closely associated with middle- and high-frequency components, whereas low-frequency responses mainly reflect coarse spatial trends [36]. When convolutional backbones repeatedly apply strided convolutions and pooling operations, these informative components may be progressively attenuated, which weakens tiny-object representation.

To quantify this effect, we analyze the spectral characteristics of tiny-object patches across backbone stages. Given an input image containing a ground-truth tiny object, we extract the corresponding region from the input image and from intermediate backbone feature maps with output strides of 4, 8, and 16. Each feature map is reduced to a 2-D activation map through channel aggregation. We then compute the 2-D Fourier transform and obtain the log amplitude spectrum. To provide a compact and rotation-invariant description of the spectral distribution, we further compute the radially averaged spectral profile

$$P(r) = \frac{1}{|\Omega_r|} \sum_{(u,v)\in\Omega_r} |F(u,v)| \tag{1}$$

where $F(u,v)$ denotes the Fourier coefficient at frequency coordinate (u,v) and Ω_r denotes the set of spectral locations with radial distance r from the spectrum center.

Fig. 1(a) shows the averaged radial spectral profiles of tiny-object regions across backbone stages. The input image

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exhibits a relatively gradual spectral decay, indicating that structural information is distributed across a broad frequency range. As feature extraction proceeds through deeper stages, the spectral curves become increasingly concentrated in the low-frequency region. Mid-frequency and high-frequency responses decrease progressively from shallow to deep features, revealing a clear low-pass filtering tendency induced by convolutional downsampling [37].

To further quantify this redistribution, we divide the normalized frequency range into three bands corresponding to low, mid, and high frequencies. Fig. 1(b) shows the proportion of spectral energy in each band. At the input stage, spectral energy is distributed across multiple bands. As the network depth increases, low-frequency components rapidly become dominant, while mid-frequency and high-frequency contributions decrease substantially. In the deepest stage, most spectral energy is concentrated in the lowest band, indicating that the representation becomes increasingly biased toward coarse background structures.

Since tiny objects rely strongly on structural cues, such attenuation may affect them differently from background regions. To examine this effect, we compare the relative attenuation of mid-frequency and high-frequency components between tiny-object patches and randomly sampled background patches. The attenuation ratio is defined as

$$\mathcal{A}_\ell = 1 - \frac{E_\ell^{(stage)}}{E_\ell^{(input)}} \tag{2}$$

where E_ℓ denotes the spectral energy within frequency band ℓ.

As shown in Fig. 1(c), mid-frequency components exhibit consistently stronger attenuation in tiny-object regions than in background regions, especially in deeper stages. Since middle-to-high frequency components are more closely related to localized boundaries and fine structural variations [36], this result suggests that convolutional downsampling suppresses object-relevant structural information more strongly than background responses.

Although extremely high-frequency components are also attenuated, the difference between tiny-object and background regions becomes less pronounced in deeper stages because most high-frequency energy has already been heavily suppressed. Nevertheless, the persistent degradation observed in the mid-frequency band indicates that the spectral components most relevant to tiny-object structure are progressively weakened during feature extraction.

B. Distinct Roles of Amplitude and Phase in Tiny-Object Representation

A Fourier representation consists of two complementary components, namely amplitude and phase [38]. Amplitude mainly reflects the magnitude distribution of spectral responses, whereas phase determines how these responses are spatially arranged [39].

1) Amplitude–Phase Reconstruction Analysis: To visually examine the influence of amplitude and phase, we conduct a reconstruction experiment on tiny objects and background patches. For each patch, we compute its Fourier transform and construct three variants: amplitude-only reconstruction, phase-only reconstruction, and full reconstruction. In amplitude-only reconstruction, the original amplitude is retained while the phase is replaced by random values. In phase-only reconstruction, the original phase is preserved while the amplitude is normalized. Full reconstruction uses both the original amplitude and phase.

Representative examples are shown in Fig. 2. For the tiny-object patch, amplitude-only reconstruction preserves coarse intensity patterns but fails to recover the object shape and spatial arrangement, making the target difficult to recognize. In contrast, phase-only reconstruction still preserves the approximate contour and spatial layout of the object, although intensity variations become noisy. When both amplitude and phase are retained, the original patch is reconstructed accurately.

A similar tendency can be observed in background patches, although their structures are less discriminative than those of tiny objects. These examples suggest that amplitude is more related to overall spectral energy distribution, whereas phase is more closely tied to the spatial organization required to recover object structure [39].

2) Amplitude and Phase Perturbation Sensitivity Analysis: To further compare the roles of amplitude and phase, we perform a perturbation-based sensitivity analysis. Instead of perturbing a cropped patch directly, we apply controlled perturbations to the full image spectrum and then crop the same tiny-object region for evaluation. This setting better reflects how spectral corruption affects tiny-object representation in the original image.

Two perturbation types are considered. In amplitude perturbation, the amplitude spectrum is progressively attenuated in the mid- and high-frequency ranges while the original phase remains unchanged. In phase perturbation, the amplitude is preserved while phase values in the same frequency range are gradually randomized.

For quantitative evaluation, 200 tiny-object patches were randomly sampled from the training split of AI-TOD with category-unaware uniform sampling. Their structural consistency with the original patches is then measured under increasing perturbation strength. Two complementary metrics are used. The first is the structural similarity index measure, abbreviated as SSIM [40], which evaluates overall structural similarity. The second is Edge F1, which measures the preservation of edge structures extracted from the patches.

The averaged results are shown in Fig. 3. As perturbation strength increases, phase perturbation causes a faster decline in SSIM than amplitude perturbation. The difference becomes more evident in Edge F1, where phase perturbation leads to a much steeper drop in edge preservation. Under the examined perturbation setting, these results indicate that the tiny-object structure is more sensitive to phase corruption than to amplitude attenuation.

C. Implication for Frequency-Aware Representation Learning

The preceding analyses suggest two points that are directly relevant to frequency-aware representation learning. Repeated

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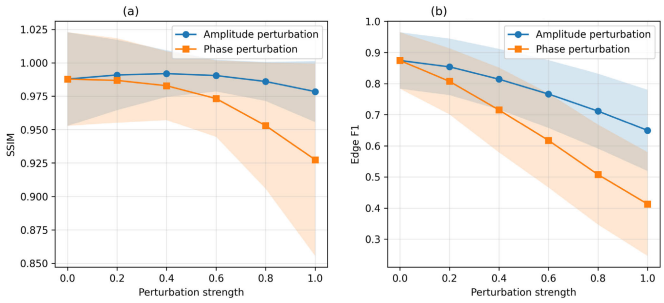


Fig. 3. Quantitative sensitivity analysis of amplitude and phase perturbations on tiny-object patches. (a) Structural similarity under increasing perturbation strength. (b) Edge structure preservation measured by Edge $F1$. Phase perturbation causes faster structural degradation than amplitude perturbation under the examined setting.

downsampling suppresses spectral components that are important for tiny-object structure, especially in the mid-frequency range. In addition, amplitude and phase contribute differently to representation quality, and phase appears more critical than amplitude alone for preserving tiny-object structure under the examined reconstruction and perturbation settings. These observations suggest that treating the Fourier spectrum as a unified entity is insufficient for this problem. A more suitable strategy is to modulate amplitude and phase separately.

IV. Methods

A. Overall Framework

Fig. 4 shows the overall architecture of the proposed detector. The framework follows a YOLO-style one-stage design with a backbone–neck–head pipeline. Different from conventional detectors that rely only on spatial-domain feature extraction, the proposed framework introduces frequency-aware representation learning into the backbone. This design is motivated by the analysis in Section III, which shows that repeated downsampling progressively suppresses spectral components that are important for tiny-object structure.

The backbone extracts hierarchical feature maps from the input image through convolutional layers. To compensate for the frequency attenuation introduced by repeated downsampling, we insert the proposed FAPR module at intermediate backbone stages. FAPR transforms feature maps into the frequency domain and applies separate modulation operators to amplitude and phase, so that spectral energy redistribution and structural preservation can be handled jointly.

Based on the recalibrated backbone features, we further employ a lightweight context modeling block derived from RepViT [41]. DS-RepViT performs efficient spatial interaction through convolutional token mixing and channel refinement, which helps aggregate spatially distributed evidence with moderate computational overhead.

The neck adopts a standard feature pyramid structure to aggregate multiscale information across spatial resolutions. High-resolution features retain fine spatial details for tiny-object localization, whereas low-resolution features provide stronger semantic abstraction.

In the detection head, we introduce FRAM for lightweight prediction refinement. FRAM uses band-limited frequency cues to guide channel and spatial attention, so that informative responses can be emphasized before final classification and localization.

B. Frequency-Aware Amplitude and Phase Recalibration Module

Based on the analysis in Section III, we introduce FAPR. The module performs frequency-domain transformation on intermediate feature maps and applies dedicated modulation operators to amplitude and phase components.

The overall workflow is illustrated in Fig. 5. FAPR consists of three main stages: frequency-domain representation, frequency-conditioned modulation generation, and inverse transformation with residual fusion.

1) Frequency-Domain Representation: Given an intermediate feature map

$$\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W} \tag{3}$$

we first transform it into the frequency domain using a 2-D Fourier transform

$$\mathbf{X}_f = \mathcal{F}(\mathbf{X}) \in \mathbb{C}^{B\times C\times H\times (W/2+1)}. \tag{4}$$

Each complex coefficient can be decomposed into amplitude and phase components

$$\mathbf{X}_f(u,v) = |\mathbf{X}_f(u,v)|e^{j\angle \mathbf{X}_f(u,v)} \tag{5}$$

where $|\mathbf{X}_f(u,v)|$ represents the spectral magnitude and $\angle\mathbf{X}_f(u,v)$ denotes the phase angle. The amplitude spectrum reflects the energy distribution across frequency bands, while the phase spectrum encodes spatial arrangement information.

2) Frequency-Aware Descriptor Construction: Instead of predicting modulation coefficients directly from feature values, FAPR generates frequency-conditioned operators from the location of each coefficient on the nonredundant Fourier grid. This design ensures that the modulation depends on intrinsic spectral coordinates rather than content-specific activations.

For each frequency coordinate (u,v), we construct a 6-D frequency descriptor

$$\mathbf{f}(u,v) \in \mathbb{R}^6. \tag{6}$$

To obtain a normalized coordinate system, we first define

$$\tilde{u} = \frac{2\min(u, H-u)}{H}, \quad \tilde{v} = \frac{2v}{W}, \quad r = \frac{1}{\sqrt{2}}\sqrt{\tilde{u}^2+\tilde{v}^2} \tag{7}$$

where $\tilde{u},\tilde{v} \in [0,1]$ are folded normalized frequency coordinates and $r \in [0,1]$ is the corresponding normalized radius.

We then construct the descriptor as

$$\mathbf{f}(u,v) = \begin{bmatrix} r \\ \sin(\pi\tilde{u}) \\ \cos(\pi\tilde{v}) \\ \frac{1}{1+\tilde{u}+\tilde{v}} \\ \frac{3r^2-1}{2} \\ 5r^3-3r \end{bmatrix}. \tag{8}$$

The six components have the following interpretations.

The first term represents normalized radial frequency and therefore reflects the overall frequency magnitude. The second and third terms encode periodic directional patterns along the two axes of the Fourier grid. The fourth term provides

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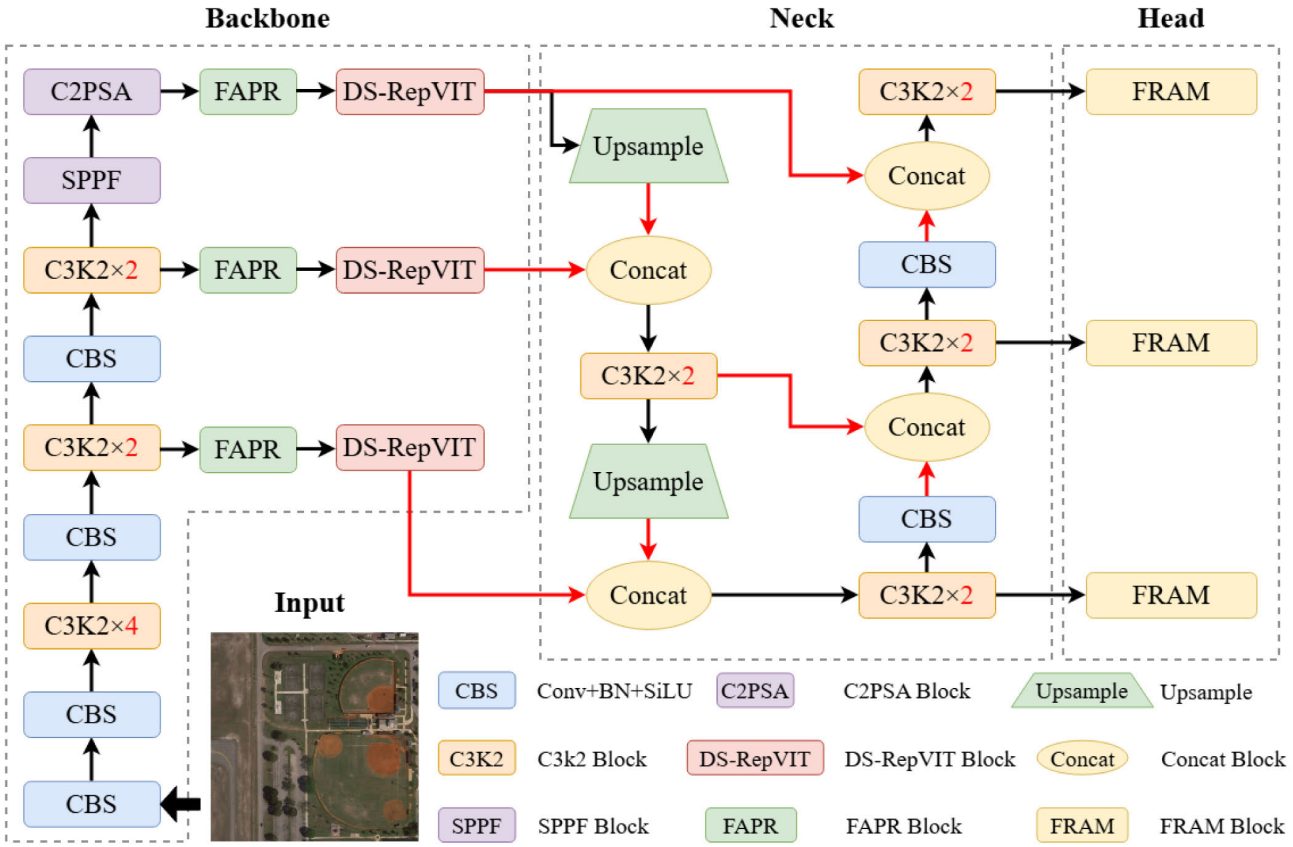


Fig. 4. Overall architecture of the proposed frequency-aware tiny-object detection framework.

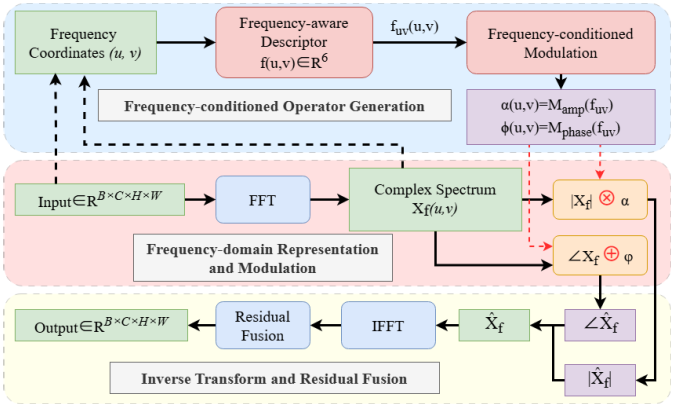


Fig. 5. Workflow of the proposed FAPR.

an inverse-distance decay cue. The fifth and sixth terms are low-order radial polynomial bases that allow the modulation functions to model nonlinear frequency responses. Overall, the descriptor captures complementary properties of a frequency location, including radial magnitude, directional periodicity, spectral decay tendency, and nonlinear radial geometry.

3) Frequency-Conditioned Modulation: The descriptor $\mathbf{f}(u, v)$ is passed through two lightweight multilayer perceptrons to generate modulation operators for amplitude and phase

$$\alpha(u, v) = \mathcal{M}_{\text{amp}}(\mathbf{f}(u, v)) \tag{9}$$

$$\phi(u, v) = \mathcal{M}_{\text{phase}}(\mathbf{f}(u, v)). \tag{10}$$

The amplitude operator $\alpha(u, v)$ is constrained by a sigmoid activation so that it acts as a positive reweighting factor for spectral magnitude. The phase operator $\phi(u, v)$ uses a hyperbolic tangent activation and is scaled to the angular range, allowing controlled phase adjustments.

The modulation is applied separately to the amplitude and phase

$$|\hat{\mathbf{X}}_f(u, v)| = |\mathbf{X}_f(u, v)| \cdot \alpha(u, v) \tag{11}$$

$$\angle\hat{\mathbf{X}}_f(u, v) = \angle\mathbf{X}_f(u, v) + \phi(u, v). \tag{12}$$

Frequency-dependent amplitude reweighting redistributes spectral energy across frequency bands, which helps reduce low-frequency dominance and relatively strengthen informative middle-frequency and high-frequency contributions after spatial-domain fusion. Phase shifting modifies the spatial arrangement encoded in the spectrum while preserving magnitude information, which helps maintain structural cues such as edges and object boundaries.

The recalibrated complex spectrum is therefore

$$\hat{\mathbf{X}}_f(u, v) = |\hat{\mathbf{X}}_f(u, v)| e^{j\angle\hat{\mathbf{X}}_f(u,v)}. \tag{13}$$

4) Inverse Transformation and Feature Fusion: The modified spectrum is transformed back to the spatial domain using the inverse Fourier transform

$$\hat{\mathbf{X}} = \mathcal{F}^{-1}(\hat{\mathbf{X}}_f). \tag{14}$$

To preserve the original representation while introducing frequency-enhanced features, we concatenate the original and recalibrated features and then fuse them with a 1×1 convolution

$$\mathbf{Y} = \text{Conv}_{1\times1}([\mathbf{X}, \hat{\mathbf{X}}]) \tag{15}$$

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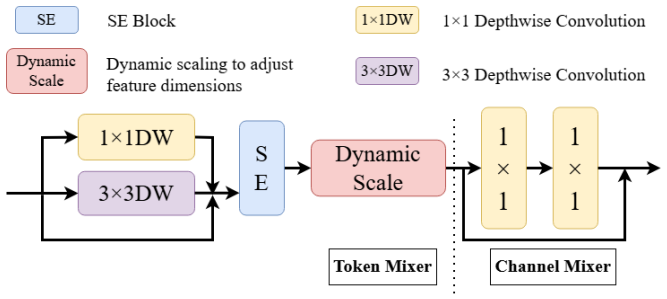


Fig. 6. Structure of the DS-RepViT module.

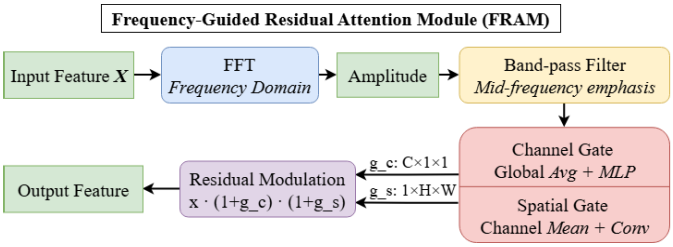


Fig. 7. Structure of the proposed FRAM.

where $[\cdot]$ denotes channel concatenation followed by a 1×1 convolution for channel alignment.

Through frequency-conditioned amplitude scaling and phase adjustment, FAPR explicitly compensates for frequency attenuation in convolutional backbones and strengthens the structural cues required for tiny-object detection.

C. DS-RepViT for Lightweight Context Modeling

While FAPR compensates for frequency degradation in backbone features, reliable detection in remote sensing scenes also benefits from contextual reasoning. Tiny objects often appear in cluttered environments, where local evidence alone may be insufficient. To enhance contextual interaction on top of the frequency-enhanced representations produced by FAPR, we introduce a lightweight context modeling module based on RepViT.

Fig. 6 shows the structure of the DS-RepViT block [41]. RepViT approximates transformer-style representation learning with efficient convolutional operations and decomposes feature transformation into a token mixer for spatial interaction and a channel mixer for channel refinement.

Given an input feature map

$$\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W} \tag{16}$$

the RepViT block performs token mixing followed by channel mixing:

$$\mathbf{Y} = \mathcal{C}\left(\mathcal{T}\left(\mathbf{X}\right)\right) \tag{17}$$

where $\mathcal{T}(\cdot)$ denotes the token mixer and $\mathcal{C}(\cdot)$ denotes the channel mixer.

The token mixer models spatial interaction using depthwise convolution

$$\mathbf{X}_{\text{token}} = \text{DWConv}\left(\mathbf{X}\right) \tag{18}$$

which aggregates local contextual information with low computational overhead.

The channel mixer further refines interchannel dependencies through a residual pointwise transformation

$$\mathbf{X}_{\text{out}} = \mathbf{X}_{\text{token}} + \text{PWConv}_2\left(\sigma\left(\text{PWConv}_1\left(\mathbf{X}_{\text{token}}\right)\right)\right) \tag{19}$$

where $\text{PWConv}(\cdot)$ denotes a 1×1 convolution and $\sigma(\cdot)$ denotes the Gaussian error linear unit activation.

To adapt the modeling capacity across backbone stages, we employ a dynamic channel expansion strategy in which the expansion ratio gradually increases with network depth. This design enables deeper stages to capture richer context while keeping early stage computation lightweight.

RepViT operates on the frequency-enhanced features produced by FAPR and serves to aggregate spatial context rather than compensate for frequency degradation itself.

D. Frequency-Guided Residual Attention in Detection Head

Although FAPR compensates for frequency degradation in backbone features and DS-RepViT further aggregates contextual evidence, prediction features may still contain weak object responses mixed with background activations. To refine these responses before final classification and localization, we introduce FRAM in the detection head. Unlike FAPR, FRAM does not directly recalibrate the full spectrum. Instead, it extracts band-limited cues to guide residual attention on detection features. The overall structure of FRAM is illustrated in Fig. 7.

Given an input detection feature map

$$\mathbf{X} \in \mathbb{R}^{B \times C \times H \times W}. \tag{20}$$

FRAM first transforms it into the frequency domain

$$\mathbf{X}_f = \mathcal{F}\left(\mathbf{X}\right) \tag{21}$$

and extracts the amplitude spectrum

$$\mathbf{A} = |\mathbf{X}_f|. \tag{22}$$

FRAM then applies a bandpass filter to retain a middle-frequency range

$$\mathbf{A}_{\text{bp}} = \mathbf{A} \odot \mathbf{M}_{\text{bp}}. \tag{23}$$

The same mask is also applied to the complex spectrum

$$\mathbf{X}_{\text{bp}} = \mathbf{X}_f \odot \mathbf{M}_{\text{bp}} \tag{24}$$

where $\mathbf{M}_{\text{bp}}$ is a binary mask defined on the frequency radius. Middle-frequency responses are used because low-frequency components are more easily dominated by coarse background trends, whereas high-frequency components are less stable. This choice is further examined in the ablation study in Section V.

Based on these filtered responses, FRAM constructs a channel gate and a spatial gate.

The channel gate models global frequency-aware channel importance by first applying global average pooling

$$\mathbf{z}_c = \text{GAP}\left(\mathbf{A}_{\text{bp}}\right) \tag{25}$$

and then using a lightweight bottleneck transformation

$$\mathbf{g}_c = \sigma\left(\text{Conv}_2\left(\delta\left(\text{Conv}_1\left(\mathbf{z}_c\right)\right)\right)\right) \tag{26}$$

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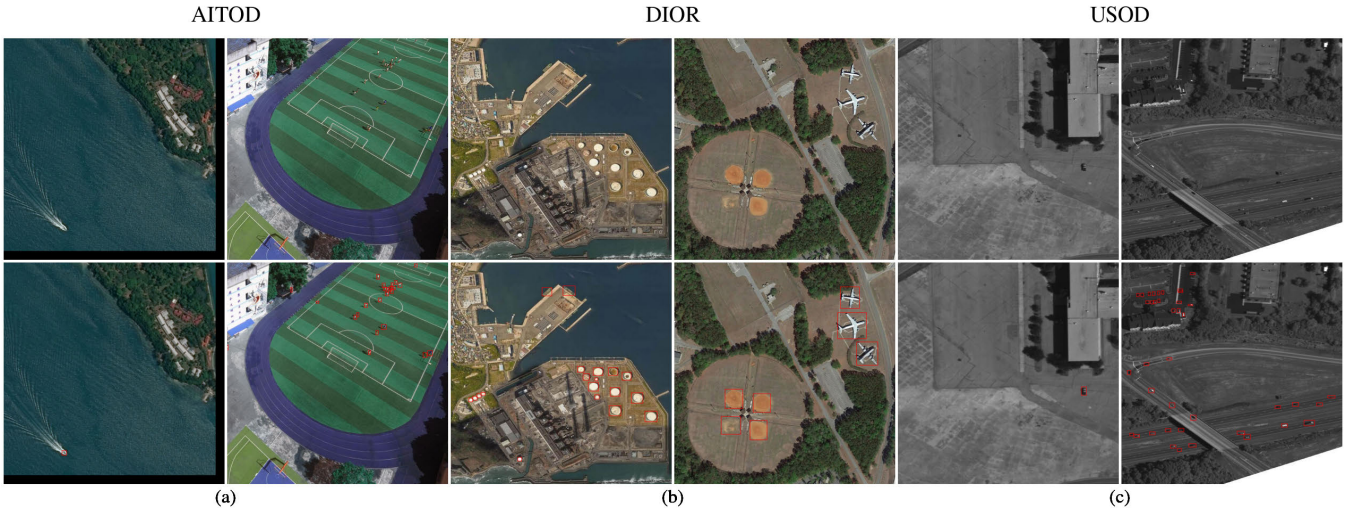


Fig. 8. Qualitative detection results of the proposed method on (a) AI-TOD, (b) DIOR, and (c) USOD. The first row shows the input images, and the second row shows the corresponding detection results.

where $\delta(\cdot)$ denotes the ReLU activation, $\sigma(\cdot)$ denotes the sigmoid activation, and $\mathbf{g}_c \in \mathbb{R}^{B\times C\times 1\times 1}$ is the resulting channel-wise gate.

The spatial gate is derived from the bandpass-filtered complex spectrum by first mapping it back to the spatial domain

$$\mathbf{H}_{\text{bp}} = \left|\mathcal{F}^{-1}\left(\mathbf{X}_{\text{bp}}\right)\right| \tag{27}$$

where $\mathbf{H}_{\text{bp}} \in \mathbb{R}^{B\times C\times H\times W}$ is a frequency-guided spatial response map.

We then average it across channels

$$\mathbf{S} = \text{Mean}_c\left(\mathbf{H}_{\text{bp}}\right) \tag{28}$$

followed by a convolutional projection

$$\mathbf{g}_s = \sigma\left(\text{Conv}_{3\times 3}\left(\mathbf{S}\right)\right) \tag{29}$$

where $\mathbf{g}_s \in \mathbb{R}^{B\times 1\times H\times W}$ denotes the spatial gate.

Finally, the two gates are applied to the original feature map in a residual form

$$\mathbf{Y} = \mathbf{X} \odot \left(1 + \mathbf{g}_c\right) \odot \left(1 + \mathbf{g}_s\right) \tag{30}$$

where $\odot$ denotes element-wise multiplication. The residual formulation preserves the original detection responses while selectively amplifying informative channels and spatial regions, which is more suitable for weak, tiny-object cues than direct suppressive modulation.

V. Experiment

A. Experimental Settings

All experiments are conducted within a YOLO11-style one-stage detection framework implemented using the Ultralytics YOLO library (v8.3.93) and PyTorch. Training is performed with distributed data parallel on two NVIDIA A6000 graphics processing units. Automatic mixed precision is disabled, and deterministic computation is enabled to improve reproducibility.

We adopt the default Ultralytics optimization configuration, which uses stochastic gradient descent with momentum. The initial learning rate is set to 0.01 and linearly decays during training, with a warm-up stage over the first three epochs. All models are trained from scratch and do not use external pretrained weights.

A unified data augmentation strategy is applied to all datasets. Mosaic augmentation is enabled, while MixUp and Copy-Paste are disabled. Hue, saturation, and value jittering, random scaling, and random translation are used. Rotation and perspective transformations are not used.

Model complexity is reported using the number of parameters and floating point operations under an input resolution of 640×640.

B. Datasets and Evaluation Metrics

1) Datasets: AI-TOD [55] is designed for tiny object detection in aerial imagery. It contains 28 036 images with annotations for 8 object categories, where most targets occupy only a few pixels. AI-TOD exhibits extremely dense object distributions and severe scale imbalance, and is used as the primary benchmark in this work.

DIOR [56] is a large-scale optical remote sensing dataset with 23 463 images and 20 object categories covering diverse scene types. Objects span a wide range of scales, from small vehicles to large infrastructures, which supports the evaluation of model generalization under complex scenarios.

USOD [18] is a small object detection benchmark constructed from visible-light imagery in UNICORN2008. It contains 3000 images with more than 43 000 annotated objects, mainly small vehicles under cluttered backgrounds and dense layouts. We follow the standard train and test split defined by the benchmark.

2) Evaluation Metrics: All datasets are evaluated using COCO-style metrics, including average precision at intersection over union thresholds from 0.50 to 0.95, denoted as AP, and average precision at 0.50 and 0.75, denoted as AP_{50} and AP_{75}.

For AI-TOD and DIOR, we additionally report size-aware metrics following the AI-TOD protocol, including AP_{vt}, AP_{t},

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TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON AI-TOD

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s	Params(M)	FLOPs(G)
Faster R-CNN [42]	11.4	27.0	8.0	0.0	8.3	23.1	–	–
FSANet [43]	22.6	52.8	15.6	7.4	21.6	29.1	31.9	125.2
HANet [44]	22.1	53.7	14.4	10.9	22.2	27.3	26.4	–
CFENet [45]	26.6	59.2	19.6	9.0	27.5	32.3	–	–
DNTR [31]	26.2	56.7	20.2	**12.8**	26.4	31.0	–	167.5
RFLA [46]	21.1	51.6	13.1	9.5	21.2	26.1	–	–
MENet [16]	23.2	56.2	15.0	9.7	23.9	25.3	41.95	148.25
NWD-RKA [34]	24.7	57.4	17.1	9.7	24.2	29.8	–	–
ORFENet [17]	24.8	55.4	18.2	9.7	24.4	28.7	–	–
FFCA-YOLO [18]	27.7	**61.7**	–	11.7	23.2	30.1	7.12	–
YOLO11s [47]	23.4	50.1	18.9	6.1	21.1	34.1	9.43	21.6
Ours	**28.7**	60.6	**25.2**	10.7	**27.9**	**37.3**	5.23	51.2

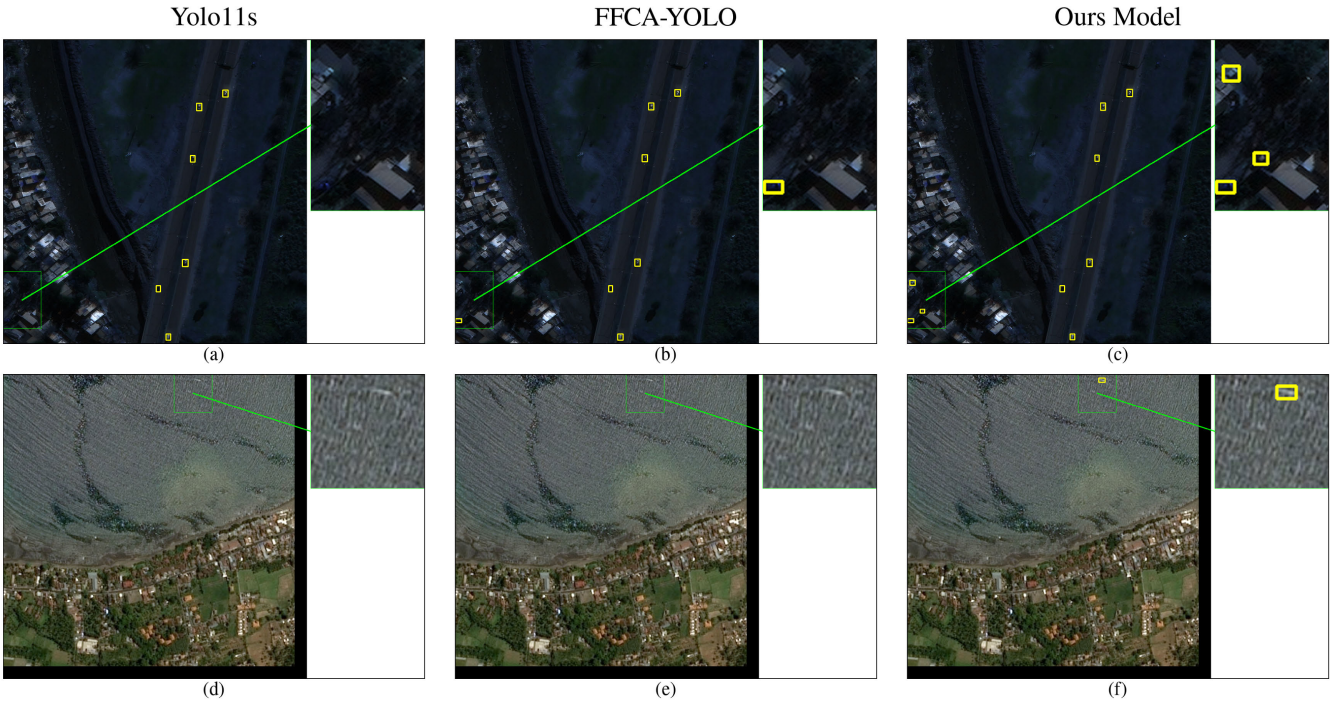


Fig. 9. Qualitative comparison between different detectors. The proposed method achieves more accurate localization and higher recall on tiny and weak objects compared with YOLO11s and FFCA-YOLO, particularly in cluttered and low-contrast scenes. (a)–(c) Detection results of YOLO11s, FFCA-YOLO, and the proposed method on Example 1. (d)–(f) Corresponding results on Example 2.

and AP_s, which correspond to very-tiny, tiny, and small object subsets, respectively. In AI-TOD, objects are partitioned by area in pixels into four ranges: very-tiny $[2, 8]$, tiny $[8, 16]$, small $[16, 32]$, and medium $[32, 64]$. Accordingly, AP_{vt}, AP_t, and AP_s are computed on the first three subsets.

For USOD, because official size-split annotations are unavailable, we report precision, recall, and $F1$-score, together with AP and AP_{50}, following common practice in prior USOD evaluations.

C. Comparison With State-of-the-Art Methods and Result Analysis

We evaluate the proposed method on three representative remote sensing benchmarks, including AI-TOD, DIOR, and USOD, and compare it with state-of-the-art detectors. Quantitative results are reported in Tables I–III. In addition, Fig. 8 visualizes qualitative detection results on the three datasets. Despite clear differences in scene layout, object density, and imaging conditions, the proposed method consistently produces accurate detections on extremely small and weak targets.

1) Results on AI-TOD: AI-TOD focuses on extremely small objects under dense aerial scenes and is highly sensitive to feature representation quality. As shown in Table I, the proposed method achieves the best overall AP of 28.7%. Compared with the YOLO11s baseline, the proposed method improves AP_{vt} from 6.1% to 10.7% and improves AP_t from 21.1% to 27.9%, demonstrating clear gains on very-tiny and tiny subsets. The improvement in AP_{75} from 18.9% to 25.2% further indicates more accurate localization under stricter

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TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR DATASET

Method	AP	AP_{50}	AP_{75}	AP_{vt}	AP_t	AP_s
Faster R-CNN [42]	32.3	55.8	32.7	0.0	1.4	9.6
Cascade R-CNN [48]	36.2	57.5	38.7	0.0	1.7	10.2
PAFPN [9]	33.5	57.2	33.9	0.0	1.4	9.6
TOOD [49]	38.4	60.3	40.9	0.0	1.9	10.3
ATSS [50]	15.3	36.6	10.2	0.0	1.0	2.9
AutoAssign [51]	37.8	63.3	38.7	0.2	3.0	11.4
DetectoRS + RFLA [46]	46.5	68.1	50.8	0.3	5.1	14.5
FCOS [52]	32.8	57.2	32.7	0.0	1.0	8.1
CenterNet [53]	33.7	54.6	35.3	0.0	0.8	6.4
CAF^2ENet-S [54]	**48.6**	72.1	52.8	**1.6**	8.0	**18.8**
YOLO11s [47]	46.5	66.6	50.0	0.3	3.8	13.8
Ours	48.5	**72.2**	**53.2**	1.1	**8.1**	15.8

TABLE III
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE USOD DATASET

Method	Precision	Recall	AP_{50}	AP
DSSD [46]	64.54	57.51	53.13	–
RefineDet [57]	88.10	82.40	85.10	–
YOLOv5n	81.01	75.32	80.34	29.81
YOLOv8n	84.12	77.14	82.52	30.11
ACDF-YOLO [58]	83.74	78.22	83.21	30.51
MSFE-YOLO [59]	82.81	77.17	83.53	30.74
SOD-YOLOv8n [60]	85.37	79.15	83.94	31.79
TPH-YOLOv5 [61]	91.00	83.70	89.50	32.10
FFCA-YOLO [18]	**92.90**	85.50	90.90	35.00
SCAF-YOLO [33]	89.60	83.60	89.30	34.30
YOLO11s [47]	88.48	82.19	88.05	33.51
Ours	91.41	**87.01**	**91.64**	**36.10**

Intersection over Union criteria. Fig. 9 provides qualitative comparisons, where the proposed method detects more tiny objects and produces tighter bounding boxes, particularly in cluttered and low-contrast regions.

2) Results on DIOR: DIOR provides a more diverse evaluation scenario with broader category coverage and more complex scene composition. As reported in Table II, the proposed method achieves the best AP_{50} and AP_{75} and remains competitive in AP. Compared with YOLO11s, AP increases from 46.5% to 48.5%, while AP_{vt} increases from 0.3% to 1.1% and AP_t increases from 3.8% to 8.1%. These results indicate that the proposed frequency-enhanced representation learning generalizes beyond ultradense tiny-object benchmarks and remains effective under diverse scales and complex backgrounds. The improvement in AP_{75} from 50.0% to 53.2% further confirms enhanced localization accuracy.

3) Results on USOD: USOD represents an extremely small object detection scenario dominated by tiny vehicles under dense layouts. As shown in Table III, the proposed method achieves the best AP and AP_{50} among all compared detectors. Compared with YOLO11s, AP increases from 33.51% to 36.10%, and AP_{50} increases from 88.05% to 91.64%. The proposed method also substantially improves recall from 82.19% to 87.01%, while maintaining competitive precision. These results suggest that the proposed framework remains robust when object sizes approach the lower detection limit.

4) Overall Discussion: Across all three datasets, the proposed method consistently improves performance on tiny-object subsets and under stricter localization metrics. The gains in AP_{vt} and AP_t support the motivation that frequency-aware backbone enhancement mitigates frequency attenuation introduced by convolutional downsampling. Meanwhile, the consistent improvements in AP_{75} indicate more stable and accurate localization. Overall, jointly integrating frequency-domain enhancement in the backbone, lightweight context modeling, and frequency-guided refinement in the head leads to robust and generalizable gains for remote sensing small object detection.

D. Ablation Studies

We conduct ablation studies on the USOD dataset to quantify the individual contributions and coupling effects of the proposed components, including the FAPR, DS-RepViT, and the FRAM. We report AP, AP_{50}, and AP_{75}, together with precision, recall, and $F1$-score. Model complexity is reported in terms of parameters and floating point operations under an input resolution of 640×640. In addition, inference efficiency is evaluated using single-GPU, batch-1 runtime statistics.

1) Factorial Ablation of Framework Components: Table IV reports a factorial ablation with all 2^3 combinations of FAPR, DS-RepViT, and FRAM.

Backbone Enhancement and Lightweight Context Modeling: Enabling DS-RepViT alone improves AP from 34.64 to 34.95, with AP_{75} increasing from 14.20 to 14.79. Enabling FAPR alone improves AP to 35.08 and increases recall from 86.60 to 87.35, indicating that frequency-aware recalibration strengthens object-sensitive cues. When combining FAPR and DS-RepViT, AP further increases to 35.46, and AP_{75} increases to 15.45, suggesting that frequency compensation and lightweight context aggregation provide complementary benefits for both detection accuracy and strict localization.

Head-Level Refinement and Module Coupling: Enabling FRAM alone improves AP to 34.95 while keeping the complexity unchanged, indicating that frequency-guided residual

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TABLE IV

FACTORIAL ABLATION ON THE USOD DATASET. AP DENOTES $AP_{50:95}$. ΔAP INDICATES THE GAIN OVER THE BASELINE

FAPR	DS-RepViT	FRAM	AP	ΔAP	AP_{50}	AP_{75}	P	R	F1	Params(M)	GFLOPs
×	×	×	34.64	+0.00	90.38	14.20	90.15	86.60	0.8834	4.43	36.2
×	×	✓	34.95	+0.31	91.01	14.52	91.34	87.36	0.8931	4.43	36.2
×	✓	×	34.95	+0.31	90.71	14.79	90.86	85.37	0.8803	4.96	45.7
×	✓	✓	33.89	-0.75	89.12	13.81	89.95	84.34	0.8705	4.96	45.7
✓	×	×	35.08	+0.44	91.60	14.28	90.45	87.35	0.8885	4.69	41.7
✓	×	✓	34.48	-0.16	89.56	14.92	90.32	85.68	0.8794	4.70	41.8
✓	✓	×	35.46	+0.82	91.27	15.45	90.70	87.64	0.8914	5.22	51.2
✓	✓	✓	**36.10**	**+1.46**	**91.64**	**15.21**	**91.41**	**87.01**	**0.8916**	5.23	51.2

TABLE V

COMPLEXITY AND INFERENCE EFFICIENCY ANALYSIS ON THE USOD DATASET. LATENCY IS REPORTED IN MILLISECONDS PER IMAGE. END-TO-END LATENCY INCLUDES PREPROCESSING, FORWARD INFERENCE, AND POSTPROCESSING. ALL RUNTIME STATISTICS ARE MEASURED ON A SINGLE NVIDIA RTX A6000 GPU WITH AN INPUT RESOLUTION OF 640 × 640 AND BATCH SIZE 1

Model Variant	AP	Params (M)	GFLOPs	Inference (ms)	End-to-End (ms)	Inference FPS	End-to-End FPS
YOLO11s	33.51	9.43	21.6	7.50	12.89	133.4	77.6
YOLO11m	34.72	20.10	68.3	9.71	15.19	103.0	65.9
None	34.64	4.43	36.2	9.66	15.30	103.5	65.4
+FRAM	34.95	4.43	36.2	11.16	17.11	89.6	58.4
+DS-RepViT	34.95	4.96	45.7	12.00	17.62	83.3	56.8
+FAPR	35.08	4.69	41.7	14.33	20.39	69.8	49.0
ALL	**36.10**	5.23	51.2	17.20	22.97	58.1	43.5

attention introduces negligible additional overhead relative to the corresponding backbone configuration. However, directly combining DS-RepViT and FRAM without FAPR leads to a performance drop, with AP decreasing to 33.89. A similar trend is observed when enabling FAPR and FRAM without DS-RepViT, where AP becomes 34.48. This suggests that FRAM benefits from sufficiently enhanced backbone representations and is most effective when used together with frequency-recalibrated and context-enhanced features.

Full Model: When all components are enabled, the full framework achieves the best overall performance, reaching 36.10 AP and 91.64 AP_{50}, and improving recall to 87.01 with an $F1$-score of 0.8916. Notably, the gain in strict localization remains consistent, with AP_{75} reaching 15.21. These results confirm that backbone-level frequency recalibration, lightweight context aggregation, and head-level frequency-guided refinement provide complementary benefits and are most effective when jointly integrated.

2) Complexity and Inference Efficiency Analysis: Table V reports the complexity and inference efficiency of different model variants. We additionally include YOLO11s and YOLO11m as standard reference models. Note that "None" denotes our customized YOLO11-style detector with all proposed components removed, which serves as the internal baseline for module-level ablation.

Module-Level Runtime Overhead: Compared with None, FRAM introduces a moderate latency increase while keeping the parameter count and GFLOPs unchanged, which is consistent with its lightweight head-level refinement role. DS-RepViT further reduces inference speed due to the added context aggregation operations. FAPR causes a relatively larger latency increase, although its computational complexity remains moderate, which is mainly attributed to the lower hardware efficiency of FFT-based operations compared with standard convolutional operators.

Overall Efficiency of the Full Model: The full model achieves the highest AP of 36.10 while running at 17.20 ms per image in forward inference and 22.97 ms per image end-to-end, corresponding to 58.1 FPS and 43.5 FPS, respectively. These results indicate that the proposed detector maintains practical inference efficiency while achieving the best detection accuracy among the ablation variants.

Comparison With Standard YOLO Baselines: Compared with the larger YOLO11m reference model, the proposed full model achieves higher AP with substantially fewer parameters, although at a higher runtime cost. This observation suggests that the performance gain is not merely obtained by scaling model capacity, but by introducing frequency-aware recalibration and coordinated refinement tailored for tiny-object detection.

3) Internal Ablation of FAPR Under the Full Framework: Table VI analyzes FAPR design choices under the full framework where DS-RepViT and FRAM are enabled. The "w/o FAPR" setting refers to disabling FAPR while keeping DS-RepViT and FRAM activated, which explains why its parameters and GFLOPs are consistent with the corresponding configuration without FAPR.

Amplitude and Phase Modulation: Phase-only modulation outperforms amplitude-only modulation, improving AP from 33.89 to 34.85, whereas amplitude-only modulation degrades performance to 32.56. This observation is consistent with the frequency mechanism analysis in Section III, where phase

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TABLE VI

ABLATION STUDIES OF FAPR ON THE USOD DATASET. UNLESS OTHERWISE SPECIFIED, DS-REPVIT AND FRAM ARE ENABLED IN ALL SETTINGS, AND WE ONLY SWITCH FAPR ON/OFF OR ALTER ITS INTERNAL DESIGN

Variant	Freq	$\lvert X\rvert$	$\angle X$	Desc	AP	AP_{50}	AP_{75}	P	R	F1	Params(M)	GFLOPs
A. Amplitude and phase modulation												
w/o FAPR	×	×	×	–	33.89	89.12	13.81	89.95	84.34	0.8705	4.96	45.7
FAPR-Amp	✓	✓	×	6D	32.56	86.28	13.22	88.13	79.57	0.8363	5.23	51.2
FAPR-Phase	✓	×	✓	6D	34.85	90.20	14.45	90.42	85.27	0.8777	5.23	51.2
FAPR (Amp+Phase)	✓	✓	✓	6D	**36.10**	**91.64**	**15.21**	**91.41**	**87.01**	**0.8916**	5.23	51.2
B. Frequency descriptor design (Amp+Phase enabled)												
FAPR-Radial	✓	✓	✓	R	34.98	90.80	14.87	91.46	86.45	0.8888	5.23	51.2
FAPR-Scalar	✓	✓	✓	S	35.39	90.94	15.54	91.45	86.11	0.8870	5.20	50.7
FAPR-6D (Ours)	✓	✓	✓	6D	**36.10**	**91.64**	**15.21**	**91.41**	**87.01**	**0.8916**	5.23	51.2

Freq: FFT and inverse FFT. **Desc**: R denotes radial-only encoding, S denotes scalar descriptor, and 6D denotes the proposed descriptor.

TABLE VII

EFFECT OF FREQUENCY BAND SELECTION IN THE FRAM ON USOD UNDER THE FULL FRAMEWORK

Band setting	P	R	F1	AP_{50}	AP_{75}	AP
Low	**0.9153**	0.8660	0.8900	91.42	14.59	35.16
Mid	0.9141	**0.8701**	**0.8916**	**91.64**	**15.21**	**36.10**
High	0.8917	0.8092	0.8484	87.54	12.43	32.62
All	0.9022	0.8509	0.8758	89.69	13.32	33.78

is more critical for preserving tiny-object structures. When jointly modulating amplitude and phase, FAPR achieves the best performance, reaching 36.10 AP and improving AP_{75} to 15.21, indicating that amplitude redistribution becomes beneficial when structural integrity is preserved by phase-aware adjustment.

Frequency Descriptor Design: Replacing the proposed six-dimensional descriptor with a radial-only encoding reduces performance to 34.98 AP. The scalar descriptor improves AP to 35.39 but remains inferior to the full descriptor. The full six-dimensional descriptor achieves the best results, suggesting that richer frequency coordinates enable more effective frequency-conditioned modulation. The complexity remains nearly unchanged across descriptor variants, while the scalar variant slightly reduces parameters and GFLOPs due to its simplified conditioning mechanism.

4) Effect of Frequency Band Selection in FRAM: FRAM is intended as a lightweight prediction-stage refinement module that complements backbone-level frequency compensation, rather than as a standalone major source of improvement. Since its guidance signal is derived from a bandpass filtered amplitude map, we further examine the effect of frequency band selection under the full framework on USOD. Specifically, four settings are considered, including low-frequency, middle-frequency, high-frequency, and full-spectrum guidance. The results are reported in Table VII.

The middle-frequency setting yields the best overall performance, achieving 36.10 AP, 91.64 AP_{50}, and 15.21 AP_{75}, together with the highest recall and $F1$-score. The low-frequency variant remains competitive but is consistently inferior in AP and AP_{75}, while the full-spectrum variant shows a more noticeable performance drop. The high-frequency variant performs the worst among all compared settings. These results indicate that FRAM is sensitive to the choice of frequency guidance and is most effective when the refinement is driven by middle-frequency responses.

This trend is consistent with the analysis in Section III, where repeated downsampling is shown to suppress structurally informative components of tiny objects. Compared with low-frequency responses, which are more easily dominated by coarse background trends, and high-frequency responses, which are less stable, middle-frequency statistics provide a more suitable guidance signal for prediction-stage refinement. Overall, this experiment supports the use of middle-frequency guidance in FRAM and further confirms its role as a complementary refinement module in the full framework.

E. Visual Interpretation of FAPR

Fig. 10 shows the channel-averaged feature maps before and after FAPR, together with magnified regions centered on tiny objects. Before FAPR, responses are relatively diffuse and are often mixed with background activations. After FAPR, responses become more concentrated around target locations, while background responses are comparatively suppressed. This visual change indicates improved signal-to-background contrast in tiny-object representation.

The spatial change is consistent with the frequency-domain evidence in Fig. 11. Compared with the pre-FAPR spectrum, the post-FAPR spectrum shows reduced energy concentration near the origin and relatively stronger responses at larger radii. The difference map further highlights suppression in low-frequency bands and enhancement in middle-to-high-frequency bands. This pattern is consistent with the role of FAPR in alleviating low-frequency dominance in backbone features.

Fig. 12 provides a compact summary using radially averaged spectra. The post-FAPR curve is lower in the low-frequency region and higher in the middle-to-high-frequency region than the pre-FAPR curve, which further verifies the spectral rebalancing effect. Together, these visualizations show that FAPR performs structured frequency-aware recalibration instead of uniform feature amplification.

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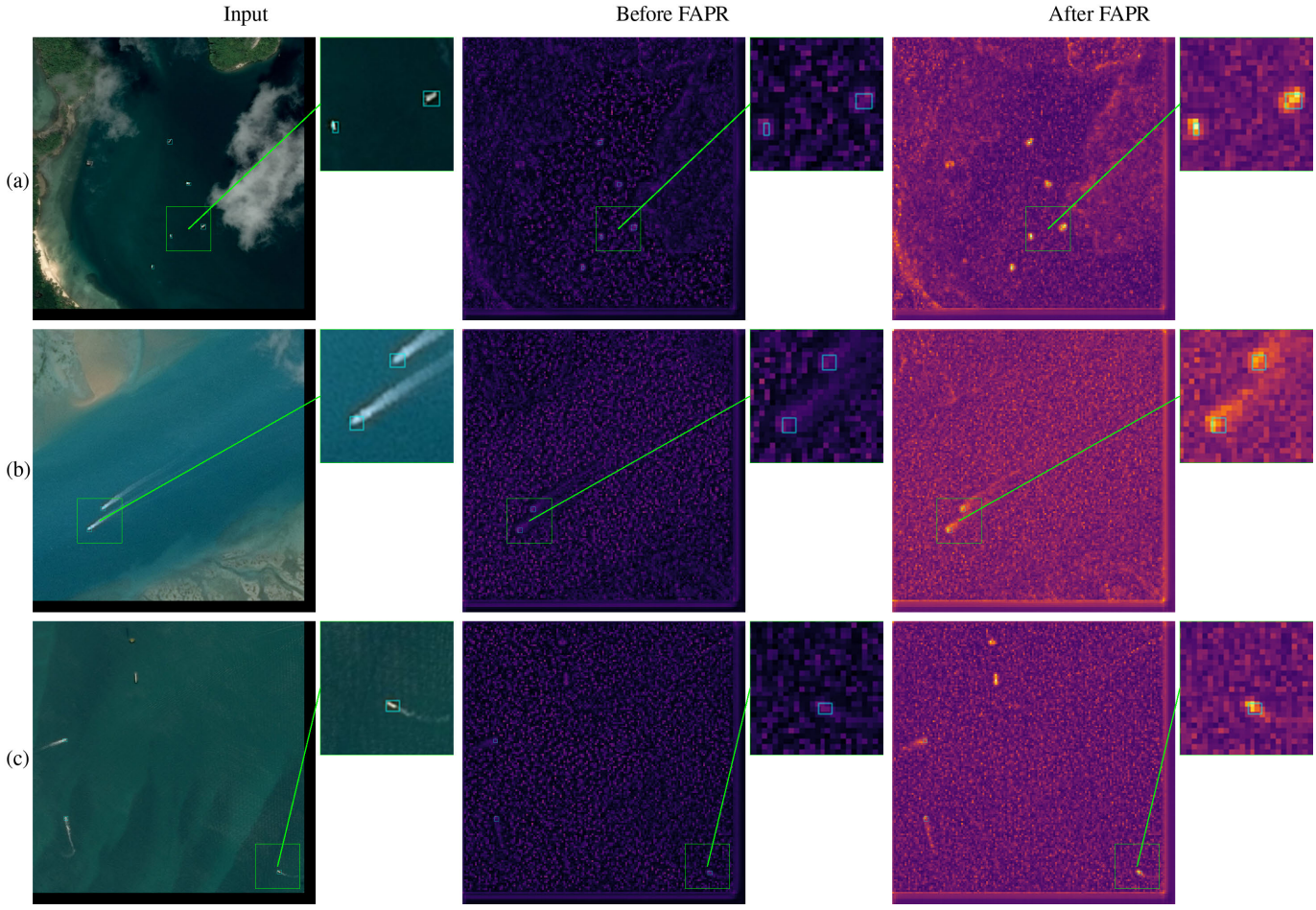


Fig. 10. Spatial visualization of FAPR. (Left to right) Input images, feature maps before FAPR, and feature maps after FAPR. Highlighted regions and magnified patches show that FAPR produces more concentrated responses around tiny objects and suppresses background-dominated activations. (a)–(c) Three representative examples from AI-TOD, illustrating tiny-object responses before and after FAPR.

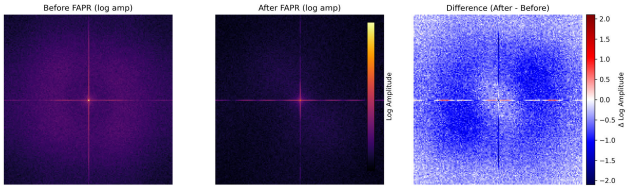


Fig. 11. Frequency-domain analysis of FAPR. Log-amplitude spectra of feature maps before FAPR (left), after FAPR (middle), and their difference (right). The difference map indicates reduced low-frequency dominance and relatively enhanced middle-to-high-frequency responses.

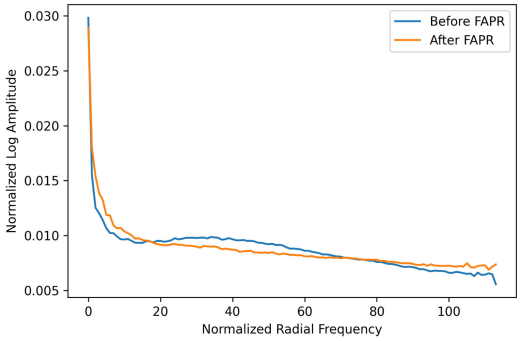


Fig. 12. Radially averaged spectra of feature maps before and after FAPR. The post-FAPR curve indicates reduced low-frequency dominance and increased middle-to-high-frequency responses.

F. Failure Cases and Limitations

Although the proposed detector improves tiny-object representation through frequency-aware recalibration and prediction-stage refinement, several challenging scenarios remain difficult. To better understand the remaining limitations, we further analyze representative failure cases on the DIOR dataset. As shown in Fig. 13, the residual errors are mainly associated with extremely small targets, dense repetitive layouts, severe intrascene scale imbalance, and cluttered background textures.

Fig. 13(a) shows a cluttered urban scene in which many extremely small objects remain undetected. In addition, some background structures with similar local appearance patterns may still trigger false responses. This suggests that when target signals are already very weak, and the surrounding textures are highly complex, frequency-aware enhancement can only strengthen the residual discriminative cues but cannot reliably recover targets whose spatial support is extremely limited.

Fig. 13(b) presents a scene with severe intrascene scale imbalance. Although visually salient large objects can still be localized, many much smaller instances remain difficult to detect. Moreover, weak roadside shadows or similar thin

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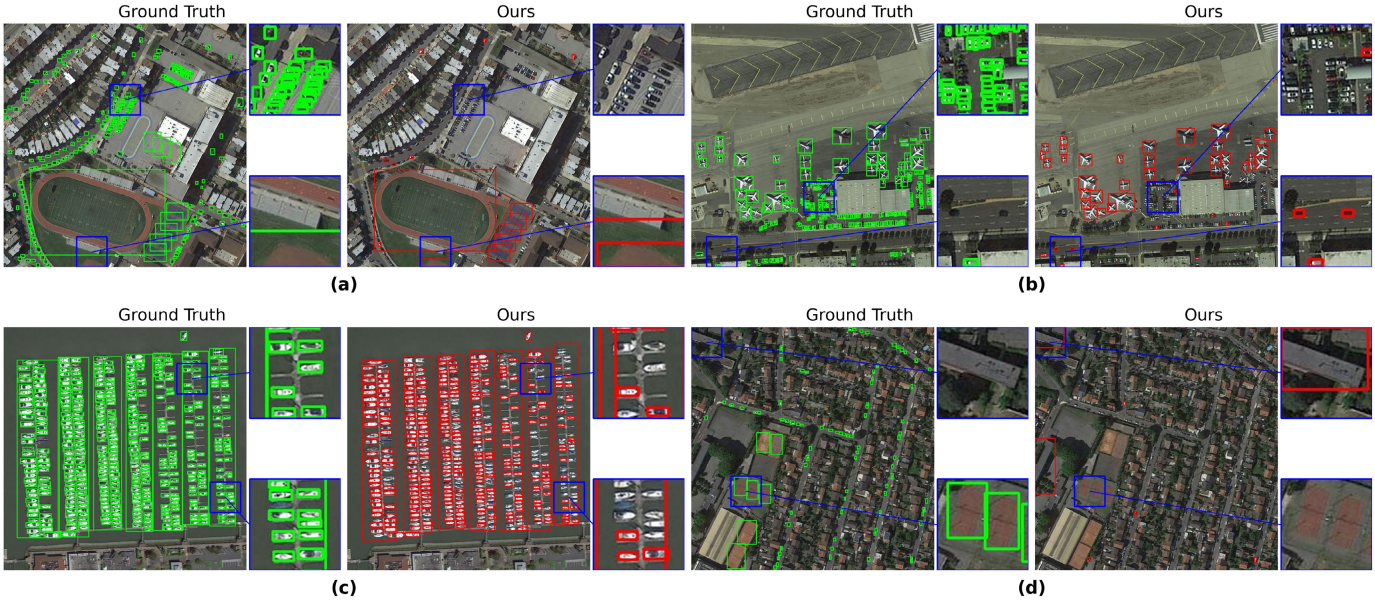


Fig. 13. Representative failure cases on the DIOR dataset. Each pair shows the ground truth and the corresponding predictions of the proposed method. The examples mainly illustrate missed detections of extremely small targets, residual errors in dense repetitive layouts, severe intrascene scale imbalance, and confusion caused by cluttered background textures. Green boxes denote ground truth, and red boxes denote predictions of the proposed method. (a) Cluttered urban scene with extremely small undetected objects. (b) Severe intrascene scale imbalance. (c) Dense repetitive layout. (d) Strong background texture interference.

textures may occasionally be confused with tiny targets. This indicates that, beyond frequency degradation, strong scale imbalance and ambiguous local structures continue to limit tiny-object recognition.

Fig. 13(c) illustrates a dense repetitive layout. The detector successfully recovers a considerable portion of the targets, but still misses some adjacent instances in highly crowded regions. This observation suggests that dense instance separation remains challenging even when the overall object category is visually recognizable, since many neighboring targets share similar local structures and appear with very small interinstance spacing.

Fig. 13(d) shows a more difficult case in which weak targets are completely missed under strong background texture interference. In this case, the target appearance is highly ambiguous relative to the surrounding man-made structures, which may still induce false responses while the true targets are not detected. This reflects an intrinsic limitation when the available spatial support is extremely limited, and the background exhibits similar structural patterns.

Overall, these examples indicate that the proposed method mainly alleviates downsampling-induced frequency degradation but does not fully resolve challenges related to extremely weak target signals, dense instance separation, severe scale imbalance, or background texture confusion. Future work may investigate finer-resolution prediction heads, density-aware localization strategies, and more adaptive frequency-region modeling for extremely small targets under cluttered remote sensing scenes.

VI. Conclusion

This article studied remote sensing tiny-object detection from a frequency-aware perspective. We showed that repeated convolutional downsampling progressively biases feature representations toward low-frequency responses and weakens structurally informative components that are important for tiny-object recognition and localization. The analysis further indicated that phase appears more critical than amplitude alone for preserving tiny-object structure under the examined settings, which motivates decoupled frequency-aware modeling.

Based on this observation, we proposed a frequency-aware detection framework built on a YOLO11-style one-stage detector. Its core component is the proposed FAPR, which performs frequency-conditioned decoupled modulation on amplitude and phase to compensate for downsampling-induced spectral bias. In addition, DS-RepViT and the FRAM were incorporated as lightweight components for context aggregation and prediction refinement.

Experiments on AI-TOD, DIOR, and USOD showed consistent improvements over strong baselines. The proposed method achieved clear gains on very-tiny and tiny subsets, together with improved localization under stricter evaluation metrics. Ablation studies and visual analysis further supported the effectiveness of the proposed design, especially the role of frequency-aware amplitude-phase compensation in improving tiny-object representation.

Future work will focus on reducing the computational overhead of frequency-aware modeling and extending the framework to broader sensing conditions.

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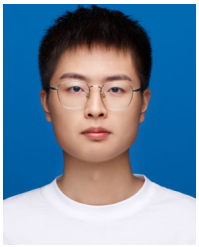
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